



FARIZ ACHMAD FAIZAL

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PROFESSIONAL SUMMARY

Electrical Engineer specializing in Control Systems, IoT, and Industrial Automation with professional experience in airport infrastructure development and technology startup leadership.

Currently working as an Electronic Officer responsible for Quality Assurance in the development project of Sultan Hasanuddin International Airport, supervising implementation of critical electronic systems including PLC/SCADA, fire alarm systems, baggage handling systems, public address systems, and data network infrastructure.

Co-Founder and Chief Technology Officer of a technology startup focusing on automation, IoT systems, and software platforms. Experienced in product R&D, system architecture design, embedded development, backend engineering, and production server deployment.

Strong technical background in embedded systems, industrial instrumentation, IoT platforms, and Linux-based infrastructure.

WORK EXPERIENCE

Electronic Officer – Quality Assurance (Airport Development Project)

PT Angkasa Pura I – Sultan Hasanuddin International Airport

Makassar, Indonesia

Nov 2023 – Present

Responsible for supervising implementation and quality assurance of electronic systems in the airport expansion project.

Key responsibilities:

- Supervising installation and commissioning of airport electronic infrastructure.
- Ensuring quality compliance of systems including:

- Data Network Infrastructure
 - PLC & SCADA for Baggage Handling System (BHS)
 - Fire Alarm System (Notifier)
 - Public Address System (TOA, Bosch)
 - X-Ray Hold Baggage Screening (Leidos)
 - Coordinating with contractors and engineering teams during system integration and testing.
 - Supporting configuration and troubleshooting of Cisco networking systems.
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Chief Technology Officer & Co-Founder

PT Media Teknologi Celebes
Makassar, Indonesia
2023 – Present

Co-founded a technology startup specializing in **automation, IoT solutions, and software development**, with primary applications in aquaculture technology and industrial monitoring systems.

Key responsibilities:

- Leading **research and development (R&D)** of company products and client solutions.
 - Designing **system architecture for IoT and automation platforms**.
 - Preparing **Product Requirement Documents (PRD)** and technical system specifications.
 - Coordinating multidisciplinary engineering teams including embedded engineers and backend developers.
 - Developing embedded systems using **Arduino, ESP32, and industrial sensors**.
 - Developing backend services using **Node.js, Django, and FastAPI**.
 - Managing server deployment and infrastructure using **Linux, Docker, Nginx, and cloud networking**.
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Intern – Air Traffic Navigation Services

AirNav Indonesia (MATSC Makassar)

Jan 2022 – Mar 2022

- Assisted maintenance of national air navigation systems.
 - Developed a web-based monitoring system for server room temperature.
 - Gained operational exposure to aviation communication and navigation infrastructure.
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EDUCATION

Bachelor of Electrical Engineering

Universitas Hasanuddin

Specialization: Control Systems and Instrumentation

GPA: 3.79 / 4.00

2019 – 2023

TECHNICAL SKILLS

Programming

Python | C++ | JavaScript

Backend Development

Node.js | Django | FastAPI

Embedded Systems & IoT

Arduino | ESP32 | Raspberry Pi | STM32

Sensor Integration

Industrial Instrumentation

Industrial Automation

PLC Programming (Allen-Bradley, Siemens)

SCADA Systems

Control Systems Engineering

Field Instrumentation

Infrastructure & Deployment

Linux Server (Ubuntu / Debian)
Docker | Kubernetes (Basic)
Nginx | VPN & Cloudflare Tunnel

Monitoring Systems

Grafana | Prometheus

SELECTED PROJECTS

Aquanotes – IoT Aquaculture Monitoring System

Developed an IoT platform for monitoring shrimp pond water quality including temperature, TDS, dissolved oxygen, pH, ammonia, and environmental parameters.

Bioassay Color Mapping – Plankton Density Detection

RGB-based monitoring system using TCS34725 sensor and camera with KNN classification for plankton density detection.

Ergoquipt – Reaction Time Measurement System

Multi-device reaction testing system using ESP32 and Arduino with real-time web interface and millisecond-level measurement.

Y-BOT – Autonomous Mobile Robot

SLAM-based mobile robot using Kinect camera and ROS for mapping and navigation.

ACHIEVEMENTS

Research Grant – Indonesian Student Creativity Program (2021, 2022)
National Finalist – Asia-Pacific Broadcasting Union Robot Contest (2021)
2nd Place – Scientific Paper Competition, AirNav Makassar (2022)
Silver Medal – LKTI Futura, Universitas Padjadjaran (2022)
Top 100 – Pertamina Foundation PFMuda Competition (2022)

LANGUAGES

Bahasa Indonesia – Native
English – Professional Working Proficiency